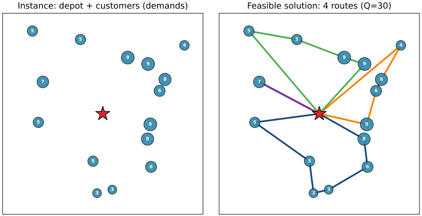
Learning-Guided Simulated Annealing for the Capacitated Vehicle Routing Problem

ICAPS 2026

Jules Andretti, Jérémie Cabessa, Yann Strozecki

David Laboratory, UVSQ – Université Paris-Saclay

2026



Objective

Minimize total travel distance

$$\min \sum_{(i,j)\in E} c_{ij}\, x_{ij}$$

Constraints

- N customers, each visited exactly once
- Route load $\leq Q$ (vehicle capacity)

Our Goal: Accelerate an Improvement Metaheuristic

Metaheuristics on CVRP

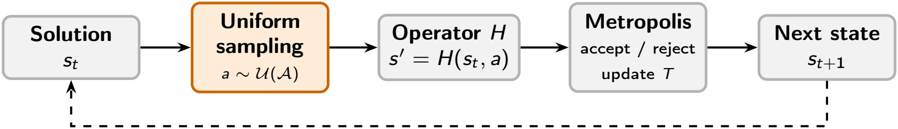
- ✓ Robust & simple
- ✓ Handle hard constraints (capacity) naturally
- × Moves proposed **at random**
  ⇒ blind, **slow** convergence

Our idea

Keep the metaheuristic — replace **only the blind randomness** with a **tiny model** (small MLP, **no heavy network**).

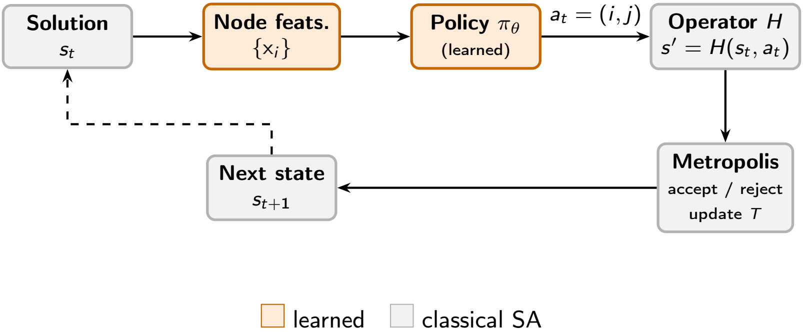
Extends Neural SA [Correia et al., 2023] from TSP to ***CVRP****.*

Simulated Annealing (SA)



- SA proposes moves **blindly**.
- We change **one** block — the proposal — and make it **guided**.

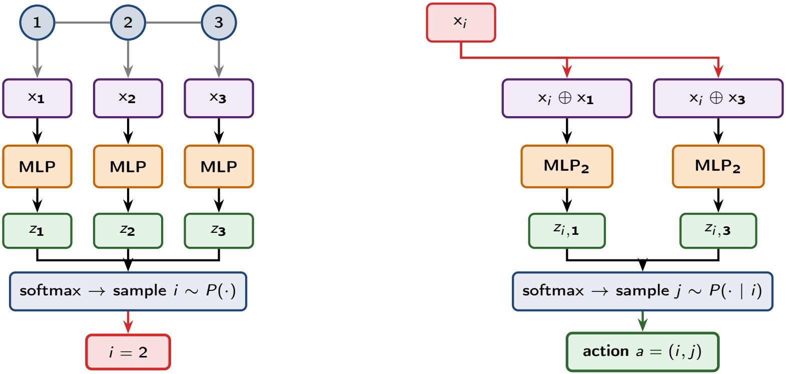
Learning-Guided Simulated Annealing (LG-SA)



How the Pair (i, j) is Sampled

Stage 1 – pick i

Stage 2 – pick j given i

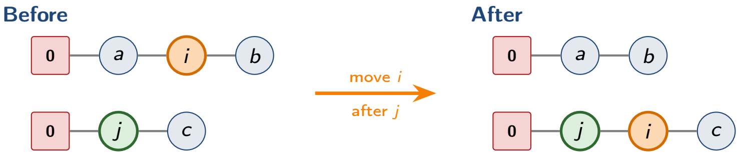


Conditional factorization: $P(i,j) = P(i) \cdot P(j \mid i)$ $O(N)$ **per step**.

Neighborhood Operator H

Three standard moves – each takes a pair (i, j)

- Swap: exchange i, j
- **Insertion**: remove i, reinsert after j
- 2-opt: reverse $i \to j$



*We fix **insertion**: best fit with capacity — many feasible inter-route moves.*

Ensuring Feasible Neighbors

Proactive masking (ours)

Reuse the conditional sampling:

- Pick i *freely* from $\mathcal{C}$ – no capacity check.
- Knowing i, compute feasible partners $\mathcal{A}_f(s, i) = \{j : H(s, (i, j)) \in \mathcal{F}\}$ and restrict $\pi_\theta(\cdot \mid s, i)$ to it.

Only $O(N)$ feasibility checks per step.

Alternatives we tested

- **Reactive rejection:** sample any a, reject if infeasible. Cheapest per step; weakest learning signal.
- **Indirect rep.:** policy on customer permutations, greedy split rebuilds feasible routes. Always feasible but $\sim 4\times$ slower per step.

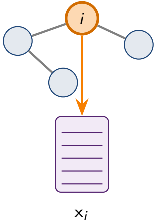
Input Features: What the Policy Sees

Spatial
- Coords, depot
- Route neighbors
- Angle, radial dist.

Capacity
- Demand q_i/Q
- Route load
- Slack, depot flag

Search
- Temperature T
- Remaining steps

Local, node-centric encoding $\Rightarrow$ **scale-invariant**: *same model from small to large N, no retraining.*



Every node i carries the **same** vector x_i — all **standardized**.

Training LG-SA with Proximal Policy Optimization (PPO)

We frame LG-SA as an **MDP** $\Rightarrow$ train with **PPO**.

State s_t

Per-node features of the current feasible solution.

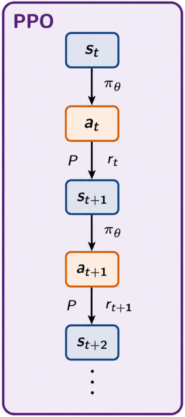
Action a_t

Node pair (i, j) for operator H.

Reward r_t – the one that works

We tested several classic RL rewards. Best: the **score difference** old $\rightarrow$ new solution ($+$ or $-$).

$= 0$ *on steps where SA rejects the move.*



PPO [Schulman et al., 2017]
actor π_θ + critic V_ϕ
(standard, off-the-shelf)

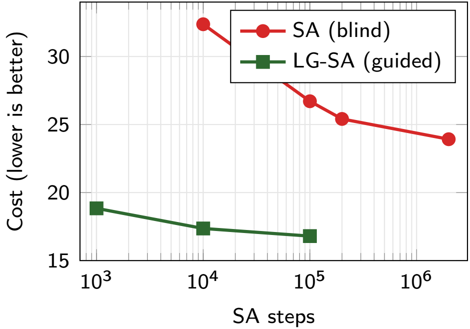
Design Choices that Matter

The method is **simple** — but a few choices are **not optional**: *remove any one and it stops working.*

- **Sample, never arg-max:** Sample from the softmax at inference.
  Greedy collapses the search; sampling gives SA the diversity it needs.
- **Keep Metropolis:** Even with learned policy, SA acceptance rule helps.
  The policy is a smarter proposer – not a replacement.
- **Multi-init training:** Mix random / NN / sweep / isolate per episode.
  A single init (esp. NN) traps the policy in deep local optima.
- **Feature choice matters:** The features are few but **carefully chosen**.
  Drop a single one and quality drops **sharply**.

Once trained, the policy reaches the same quality from any starting solution.

SA vs. LG-SA: Guidance Does the Work



The guidance does the work

- Blind SA **plateaus high** — even with a huge step budget
- Guided SA reaches **far lower cost** in a **fraction of the steps**
- Same loop, same operator — **only the proposal changed**

LG-SA vs. Classical & Learning Baselines ($N = 100$)

Method	Cost	Time
Classical		
LKH3	15.65	13 h
OR-Tools	16.61	29 m
Learning		
NLNS [Hottung & Tierney, 2022]	15.99	1 h 02 m
LIH [Wu et al., 2022]	16.03	5 h
CDCP [García-Torres et al., 2022]	15.85	19 h
Ours		
LG-SA (10^5)	**16.80**	**29 m**

A tiny model (in the sweet spot)

- **Matches** strong classical & neural methods in cost
- **Far faster** than the heavy improvement learners
- And it **scales** to large N with no retraining

Conclusion & Ongoing Work

Contributions

- **LG-SA:** PPO-guided SA that brings DRL to CVRP
- **Proactive masking:** the key that makes feasibility work
- **Conditional policy:** $O(N)$ action space
- **Scalability:** one model, $N = 50 \to 1000$, no retraining

Future directions

1. LNS operators (destroy & repair)
2. GNN encoder for global context
3. Rich VRP variants (TW, multi-depot)

Ongoing work

Same LG-SA backbone + **curriculum training** and training-side tricks

Thank you!

jules.andretti@uvsq.fr